Concurso de Projetos de Computação Avançada (6ªed) - A1 (lote A)
2025.09511.CPCA

• **Rascunho**

Período de Candidatura
10.09.2025 a 10.11.2025 - 17h00

Visão Global

Sa

Dados do IR e CO-IR

INVESTIGADOR RESPONSÁVEL

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Ficheiro obtido através do CIÊNCIAVITAE	—

Dados da Candidatura

DESCRIÇÃO GERAL

Título	Smart Models, Smarter Protocols? Benchmarking LLMs for Distributed Algorithm Design
Resumo	The verification of distributed algorithms is essential to ensure the correctness and reliability of modern distributed systems, including those used in cloud infrastructures, blockchain, and large-scale coordination services. Traditional approaches rely on formal verification tools such as theorem provers or model checkers, which, while rigorous, require specialized expertise and are often computationally expensive. The project I am working on aims to explore the alternative path of evaluating the ability of Language Models to verify distributed algorithms through natural language and algorithm's specification understanding and reasoning. To this end, I will develop a benchmark framework to assess different models on their capacity to identify algorithmic correctness and property violations, understand system assumptions (synchronous vs asynchronous, fault tolerance), and provide structured reasoning or proof-like justifications for when a property is considered incorrect. The experiments will include several representative distributed algorithms (e.g., consensus, reliable broadcast, Paxos) and their correct and faulty variants. I will design set of prompts and evaluation metrics to measure reasoning depth, consistency, and fault-tolerance awareness across models. The requested computational resources will support both the execution of large-scale experiments across multiple LLMs (including local deployment and testing) and the automated analysis of their outputs. The results will contribute to understanding whether LLMs can complement formal verification techniques and support future research on AI-assisted verification of distributed systems.
Âmbito do projeto	This project is developed in the context of a Master's thesis in Computer Science Engineering, focusing on the intersection of distributed systems, verification, and artificial intelligence. Its main objective is to design, implement, and apply a benchmarking framework that evaluates how different language models verify distributed algorithms through natural language reasoning. Its primary goal is to design and implement a benchmarking framework capable of evaluating how different language models verify distributed algorithms. Unlike traditional verification methods based on formal proofs or model checking, this benchmark will assess models' understanding through natural language, testing their ability to identify correctness properties, system assumptions, and fault-tolerance behavior. The project involves: Building a structured workload of distributed algorithms with both correct and faulty variants. Developing automated scripts to generate prompts, query models, and collect outputs across multiple configurations. Defining and applying evaluation metrics to measure detection accuracy and reasoning consistency. Running large-scale experiments with multiple language models, including hosted and locally deployed ones, comparing the different language models in terms of their efficiency and adaptability to verification tasks. The requested computational resources will enable the parallel execution of extensive experiments, efficient handling of model outputs, and reproducibility of results. The final goal is to produce an empirical foundation for understanding how language models can contribute to AI-assisted verification of distributed algorithms, paving the way for future research and publications in this emerging field.
Palavras-chave	Distributed algorithms Benchmarking Language models Verification Fault tolerance
Domínio científico	Ciências da Engenharia e Tecnologias
Área científica	Engenharia Eletrotécnica, Eletrónica e Informática
Subárea científica	Automação e Sistemas de Controlo

PLANO DE TRABALHOS COMPUTACIONAL

Tarefas, etapas e entregáveis computacionais

September 2025 – February 2026
- Literature review on benchmarking LLMs, prompt engineering, and distributed algorithms verification.
- Definition of the benchmark structure, workload and evaluation metrics.
- Selection of the language models (local and hosted) and the algorithms to be tested.
- Implementation of scripts for automated prompt generation, model querying, and data collection.
- Execution of small-scale preliminary tests to validate the benchmark framework.

March 2026 – August 2026
- Execution of large-scale tests on selected LLMs using distributed algorithms such as consensus, broadcast, and Paxos.
- Collection and analysis of outputs to evaluate correctness verification and fault-tolerance understanding.
- Refinement of automation scripts for scalability and reproducibility.
- Preparation of final datasets and benchmark report.
- Documentation and preparation of results for publication.

EQUIPA

Elementos da equipa

Nome	Instituição	Tarefa	Curriculum Vitae
Miguel Nuno Dias Alves Pupo Correia	Instituto de Engenharia de Sistemas e Computadores, Investigação e Desenvolvimento em Lisboa	Orientador	—
Diogo Lopes Vaz	Instituto de Engenharia de Sistemas e Computadores Inovação	Co-orientador	—

EXPERIÊNCIA ANTERIOR

Já utilizou plataformas de HPC ou HTC ou Virtual Research Environment (Scientific Cloud)?	Não

PROJETO CIENTÍFICO/INOVAÇÃO ASSOCIADO

Esta submissão, em caso de aprovação, contribuirá para um projeto existente?	Não

Requisitos técnicos

MODELOS COMPUTACIONAIS

Necessita recursos de computação de alto desempenho (HPC)?	Sim
Necessita de recursos de computação em Cloud científica ou VRE?	Sim

HIGH PERFORMANCE COMPUTING

Preferência de plataforma HPC	Deucalion, ARM partition Deucalion, x86 and/or GPU partition Cirrus Stratus
Número de CPU cores simultâneos necessários	0
Número total de horas de relógio necessárias	0
Total de CPU core.horas necessárias	0
RAM em GB por nó de CPU	0
Necessita utilizar GPU?	Não
Software aplicacional - HPC	—
Gráfico escalabilidade software	—
Média de CPU cores por job/trabalho computacional - HPC	0
Quantidade total de disco, em GB, necessária para suportar execuções jobs/trabalhos HPC ou armazenamento - HPC	0

CLOUD CIENTÍFICA / VIRTUAL RESEARCH ENVIRONMENTS

Número de vCPU necessários	0
Número total de horas de relógio necessárias	0
Total de vCPU.horas necessárias	0
Número de VMs (máquinas virtuais) necessárias	0
Necessita utilizar GPU?	Não
Software aplicacional - Cloud	—
Outras ferramentas e software- Cloud	—
Quantidade total de disco, em GB, necessária para suportar execuções jobs/trabalhos HTC - Cloud	0
Quantidade total de disco, em GB, por VM	0
Quantidade total de RAM, em GB, necessária- Cloud	0
Sistemas operativos	—

QUESTÕES TÉCNICAS ADICIONAIS

Eventuais requisitos especiais de suporte técnico, software ou ferramentas	—
Largura de banda da Internet antecipada (Mbit / s) necessária para o projeto computacional	—
Volume estimado de dados de Internet a ser transferido (GB) por mês	—
Locais para comunicações de Internet de alta largura de banda	—
Necessita de recursos para pós-processamento (visualização)?	—
Documento adicional	—

PEDIDO DE RECURSOS COMPUTACIONAIS

Justificação de recursos computacionais	—
Data de início	—
Data de término	—
Comentários adicionais	—
Sem prejuízo do 2º ponto do artigo 23º do regulamento aceita que as seguintes informações sejam usadas em publicações relacionadas com a RNCA?	Sim